MILESTONE 1 DELIVERABLE · SOUND SANCTUARY

Technical Audit Report

Auditoría técnica — hallazgos y recomendaciones

PREPARED FOR	PREPARED BY	SCOPE
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RESONANCIA	Sparkpath Digital	Expo SDK 54 · RN 0.81

Executive summary

This audit covers the full codebase: the audio subsystem, the database and its sync backend, the build pipeline, the Geometrix visual system, the streak system, and app-wide architecture. The purpose is to establish, from the code itself, what is production-ready and what is not, ranked by severity, so the remaining work can be planned against evidence rather than assumption.

The headline is encouraging and specific. This is a capable codebase built by a developer who solved genuinely hard problems well — a sample-accurate gapless audio engine, a rigorous server-side sync API, and correct handling of several subtle platform constraints. The problems are not incompetence. They are **incomplete consolidation**: good solutions that exist but are not fully adopted, compensated for by fragile machinery elsewhere, plus a small number of concrete defects that would affect users immediately at launch.

> **THE THROUGH-LINE**
>
> A correct, gapless audio engine exists in the code but the app does not fully trust it — two separate hand-rolled fallback systems compensate for problems the engine already solves. The same pattern of good-foundation-plus-compensating-workaround recurs across the app. The remaining work is consolidation and completion, not a rebuild.

How findings are rated

CRITICAL	Affects users at launch; fix before shipping
HIGH	Serious reliability or scaling risk; fix during the engagement
MEDIUM	Real issue, lower urgency; address in the relevant milestone
POSITIVE	Done well; noted to show what should be preserved

FINDINGS · AUDIO SUBSYSTEM

Audio architecture

El sistema de audio — la prioridad más alta.

CRITICAL

A gapless engine exists but the app runs a drift-prone fallback beside it

The project contains **bpmAudioEngine**, built on react-native-audio-api, which loops audio at sample precision with no gap and no drift. It is well written. But the mixer keeps a parallel fallback that schedules loops with a JavaScript timer against wall-clock time, selected at runtime whenever the engine is not warmed up at the moment of the first tap.

Wall-clock timers are not sample-accurate. On the fallback path, layers drift over a long session — the exact behaviour to avoid in a meditation app. Which path a session uses is decided by a cold-start race the user cannot see. The fix is to guarantee the engine is ready and remove the fallback entirely, which also deletes a large amount of code.

```
bpmSystemRef.current = bpmAudioEngine.isReady() ? "engine" : "expo";
```

CRITICAL

The same hand-rolled crossfade is duplicated in two places

The two-layer, equal-power crossfade that compensates for expo-audio's non-gapless looping is implemented twice — once in the mixer, once in the session player — each driven by its own JS interval, each carrying its own seed-and-recenter drift correction. Meanwhile bpmAudioEngine's gapless loop, which removes the need for any of it, is not used in either place.

This is the strongest single opportunity in the engagement. Routing both surfaces through the existing engine removes two fragile systems at once and materially reduces the code that has to be maintained.

HIGH

Session switching is a suppressed race, not a resolved one

The persistent main player is reused via replace() and shares one status listener, so events from a previous track can be attributed to the next. Four separate guards currently suppress the symptom — a switching flag, a skipped first tick, a skipped pre-seek tick, and manual layer resets.

The guards are carefully written, but the race remains. It should be rebuilt so session switching is deterministic rather than guarded, which removes a whole class of intermittent progress-saving and auto-resume bugs.

HIGH

The sleep timer can fail to stop audio in the background

Timer enforcement relies on JavaScript intervals, which the OS throttles when the screen is locked. The mixer reconciles this with a wall-clock check inside a status listener; the session player does not have the same reconciliation on its side, and engine-only mixes may have no expo player emitting the ticks that enforcement depends on.

For a product whose core use is falling asleep to audio, a timer that does not reliably stop playback on a locked screen is launch-relevant. Enforcement should be unified and made background-safe.

POSITIVE

The hard platform problems were solved correctly

The gapless engine, the deferred audio-session configuration that avoids an iOS launch crash, the NaN-duration lock-screen deferral, and the exclusive-focus audio mode required for lock-screen controls are all handled correctly and for the right reasons. This is the work of someone who hit the real platform walls and understood them. The audio work ahead is consolidation, not rescue.

FINDINGS · DATA LAYER

Database and synchronisation

La arquitectura de datos y la sincronización.

POSITIV
E

The schema and the sync backend are genuinely well built

The data model uses foreign keys with cascades, unique constraints, and strong Zod validation at the write boundary. The server-side sync API validates every input, authenticates every route, uses real transactions for set-replacement, and ingests play events idempotently by client event id. The durable, multi-user path is solid and should be preserved as-is.

HIGH

Social tables are stored directionally but queried undirected, and under-indexed

Friendships and direct messages store one directed row per relationship, but every real query needs both directions — a friend list, a conversation thread. The indexes cover only one direction, so the other half of each query cannot use an index and degrades toward a full-table scan as data grows.

This is invisible today and grows linearly with total data at scale — exactly the class of problem to fix before launch, not after. Shared content tables add no indexes for their feed, author, or category queries. The remedy is consistent: index for the real query patterns, and use a conversation key or a union of indexed lookups instead of an OR across columns.

MEDIUM

A denormalised like counter can drift from the truth

Shared mixes store a likes integer alongside the actual like rows in a separate table. This is a sound performance choice, but the counter and the rows can diverge unless every like and unlike updates both in a single transaction. This should be verified and a periodic reconciliation added.

MEDIUM

User-created content lives only in local storage

Mixer presets, folders, and Geometrix creations are stored solely as JSON blobs in on-device storage, with error handling that can reset the store to empty on a read failure, and no cloud backup. For content a user creates rather than consumes, this means a reinstall or a corrupt store loses their work silently. Consumption data (stats, favourites, progress) does sync; creation data should too.

MEDIUM

Third-party integrations need a security pass

Live sessions run through a Cal.com booking webhook that upserts records and a Daily.co WebRTC room. The webhook must be signature-verified and idempotent, and the book-without-account flow links by email, which is an account-linking surface to harden. Moderation for user content is currently a single hidden flag with no resolution workflow, which is thin for App Store review of a social product.

FINDINGS · STREAK SYSTEM

Streak system

El sistema de rachas.

CRITICAL

The streak shown to every user is a hardcoded test value

The home screen computes a real streak, then discards it. A debug constant overrides the result, so every user sees the same fixed number regardless of their actual activity. This is live in the current code.

This is the most user-visible defect in the audit. On the most prominent gamification surface of the app, the number is fake for everyone. It must be removed before any release. The good news: the real calculation beneath it is sound.

```
const DEV_STREAK = 3;
const currentStreakDisplay = DEV_STREAK > 0 ? DEV_STREAK : currentStreak;
```

HIGH

The described fire animation is not present on the home screen

The requested feature — a dimmed fire that lights on completing the daily goal, a navy pill expanding with the streak number, gated to once per day — is not what the home screen renders. It currently shows a static icon with a number beside it.

This confirms the streak milestone is a build, not a cleanup: the animation and the once-per-day gate need to be created, and the goal rule that lights the fire must be reconciled with the rule the streak count uses, or the two will disagree.

HIGH

The streak logic lives inside a very large single-file screen

The derivation sits inline in a home screen component well over a thousand lines that also owns search, moods, recommendations, immersive scene animation, and more. Other surfaces that display streak data may compute their own version, which is how screens end up disagreeing. The calculation should be extracted to one shared source of truth used everywhere the streak appears.

POSITIVE

The underlying streak calculation is correct

The real derivation builds a set of local-day keys, allows the streak to survive until end of day by checking today or yesterday, and walks backwards counting consecutive days. The logic is sound and timezone-aware in the right way. It simply needs to be surfaced instead of overridden, and centralised.

FINDINGS · VISUALS, BUILD, APP-WIDE

Geometrix, build pipeline, and app-wide

Rendimiento visual, pipeline de compilación y arquitectura general.

HIGH

Geometrix lag is animation running off-screen, not wrong technology

Geometrix is a clean, data-driven SVG system — a saved 'creation' is a recipe redrawn live, with shared gradient logic and no duplicated rendering code. The lag is most likely SVG node count under animation, compounded by animations that keep running on screens that are mounted but not visible. The code already contains a fix for one instance of this and notes the cause explicitly.

The remedy is to profile on device, then either reduce animated node count and pause off-screen work, or migrate the fullscreen renderer to a GPU canvas (Skia) which preserves visual quality while removing the cost. Profiling decides which, and that is the first task of that milestone.

MEDIUM

The build pipeline is sound but submission has never been configured

The EAS build profiles are correct and current — dev, preview, and production with the right distribution types. But the iOS submit credentials are still placeholder values, the Android service-account key is absent, and over-the-air updates are disabled. Store submission is genuinely unstarted; the pipeline skeleton to support it is in good shape.

HIGH

Some finished-looking screens are backed by placeholder data

At least one screen that appears complete — the in-person activities feature — is driven entirely by a hardcoded array with no data source behind it. Apple rejects apps with non-functional or placeholder features. Before launch the app needs a real-versus-placeholder inventory so every visible feature is either wired up or removed from the release.

MEDIUM

Recurring maintainability patterns

Several patterns recur and are worth addressing together: very large single-file screens that do too much; a referenced font that is not installed, so its typography silently falls back; dead code shipped in place, including disabled blocks and versioned filenames implying abandoned predecessors; and suppressed type and lint checks used as a habit. None is severe alone; together they are where future regressions will hide.

What this means

The app is closer to launch than a list of findings might suggest, and further in a few specific places. The audio, data, and sync foundations are strong; the work there is consolidation. The launch-blocking items are concrete and finite: the hardcoded streak value, the placeholder screens, the unconfigured submission pipeline, and the background timer reliability. The scaling risks — the under-indexed social tables — are invisible now and worth fixing before, not after, real users arrive.

Each of these maps to a milestone already in the plan. This report is the evidence those milestones are scoped against, and it is yours to keep regardless of what you decide next.

Blaise Sebastian
Sparkpath Digital

Gracias por la confianza. Cualquier hallazgo lo repasamos en detalle cuando quieras.